MORE: Adaptive Multi-Objective Learning for E-commerce Dialogue Systems

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Abstract

E-commerce dialogue systems must reconcile two competing objectives: responding with accurate, reasoning-grounded content while producing text that reads naturally to users. These requirements are often in tension, since responses that faithfully follow profile-based attribute logic can sound stiff, whereas fluent phrasing tends to sacrifice factual precision. Training that mixes correctness and naturalness rewards directly proves unstable, and fluency-only or reasoning-only policies each degrade the other dimension. We introduce MORE, a staged multi-objective learning framework that first trains a policy on verifiable reasoning scaffold tasks, freezes it as a reference, and then coordinates response-quality reward channels through rank-based gradient weighting. This design decouples reasoning acquisition from style optimization, allowing the frozen reference to supply reliable attribute logic while a separate policy layer adapts generation quality to the conversational context. Offline experiments on two ByteDance tasks and MultiWOZ 2.2 show that MORE improves the tested task-quality trade-off, and 14-day online traffic tests report gains in conversion, user satisfaction, and business success metrics. A human study on 241 sampled responses confirms that our automated evaluator substantially agrees with human assessment, supporting its use for scalable quality monitoring in deployment.

CCS Concepts

• **Computing methodologies → Natural language generation**.

Keywords

task-oriented dialogue, multi-objective learning, reinforcement learning, e-commerce

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1 Introduction

E-commerce dialogue is not a single optimization target but a joint problem in which the system must simultaneously reason over a structured user profile and produce a natural, fluent response. Correct responses can be read as abrupt, formulaic, or interrogative, while fluent responses can drift away from the user-supplied attributes and yield confidently incorrect suggestions. We frame the task as the coordination of profile-conditioned reasoning with open-ended generation, and we show that optimizing either criterion in isolation leaves the other unaddressed. These complementary failures show that the practical objective is a joint criterion balancing profile consistency and response naturalness. This framing motivates the design of a learning procedure that can adapt its emphasis across tasks and dialogue contexts rather than fixing a static trade-off.

Consider a customer who asks whether a loan product permits prepayment while holding a profile that specifies a particular income bracket. A correctness-only system might respond, "Prepayment is allowed for customers in your income bracket," which is factually accurate against the profile yet reads as terse and mechanical. A fluency-only system, by contrast, might produce, "Of course! You can pay off your loan early whenever you like, and we will waive any fees for you," which is natural and reassuring but contradicts the profile's restriction. Neither system satisfies the user: the first answers the query but sacrifices conversational quality, while the second reads well but asserts an unqualified allowance that the profile does not support. As Figure 1 illustrates, these two failure modes are complementary, and the practical objective is a joint criterion that enforces profile consistency and response naturalness simultaneously.

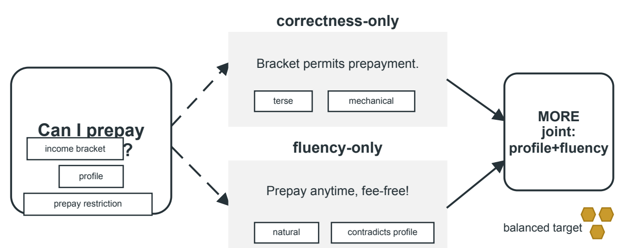

Figure 1: Correctness-only and fluency-only systems fail along complementary dimensions, as the motivating analysis shows that optimizing either criterion in isolation leaves the other unaddressed, framing the practical task as a joint criterion enforcing both.

A straightforward remedy is to combine the two criteria into a single scalar reward through a fixed weighting. In practice, however, this approach proves unstable: the training dynamics reported in Figure 2 show that a reward-only objective drives the fluency signal downward as correctness saturates, while a fixed mixed reward oscillates between the two criteria without settling on a policy that satisfies both. Neither trajectory yields a usable operating point, which indicates that static scalarization does not resolve the underlying tension. The optimization problem is therefore to coordinate multiple reward channels in a way that adapts their relative influence during training, rather than committing to a constant trade-off. This instability motivates the design of a learning procedure that re-weights rewards according to how each channel's gradient ranks across the candidate set at each update.

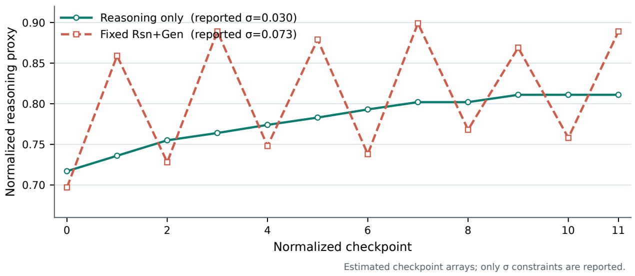

Figure 2: Training dynamics for the reward-only objective drive fluency downward as correctness saturates, while a fixed mixed reward oscillates between the two criteria without settling on a policy that satisfies both.

We introduce MORE, a staged reasoning scaffold coupled with adaptive gradient-ranked multi-reward learning, designed to resolve the optimization conflict identified above. The method proceeds in two distinct stages. In the first stage, a policy is trained jointly on two verifiable reasoning tasks: masked attribute completion, which requires the model to recover profile fields that are withheld from the input, and programmatic numerical reasoning, which requires the model to compute quantities such as eligibility thresholds or repayment terms from the structured profile. This training produces a reasoning-enhanced reference policy that is then frozen rather than updated during quality optimization. Freezing this reference decouples the verifiable objectives from subjective quality criteria, so that neither stage's gradient can interfere with the other's learning dynamics. In the second stage, response-quality optimization proceeds as follows: the frozen reference generates a candidate pool, every candidate is scored across multiple reward channels that capture distinct dimensions of response quality, and rank-based gradient weighting coordinates these channels context-sensitively. For each candidate, the training signal is shifted toward the channels where that candidate ranks most poorly, rather than applying a fixed scalar trade-off uniformly across all candidates and training steps. This per-candidate, per-channel adaptation allows the optimization to emphasize factual fidelity for some responses while emphasizing naturalness for others, depending on where each candidate falls short. The complete architecture thus replaces static scalarization with a dynamic coordination mechanism whose behavior we examine empirically in the following sections.

We evaluate MORE across three complementary settings, each scoped to the evidence it can support. On the ByteDance offline benchmark, MORE achieves joint gains on both overlap-based and human-aligned quality dimensions relative to matched baselines. On MultiWOZ, it improves task success and response diversity, while retaining the expected BLEU and CBE trade-offs. In deployment, a six-variant 14-day A/B test on the ByteDance platform measures the primary relative changes, with additional cohort and human-agent comparisons providing bounded deployment evidence. Finally, we report efficiency gains: reduced latency and shorter output lengths against explicit chain-of-thought inference. These offline, online, and efficiency contributions are each presented with their tested scope clearly delimited, so that claims of generalization remain appropriately qualified rather than overstated.

2 Related Work

Profile-conditioned dialogue systems condition generation on a static user profile, yet the attributes that govern an e-commerce exchange, such as a shopper's budget, delivery preference, or willingness to accept a substitute, shift over the course of a conversation. A fixed profile snapshot cannot capture these dynamics because it reflects a prior state rather than the evolving interaction. Consequently, such models may persist with outdated assumptions, recommending a high-end item after the user has signaled a tighter budget or failing to revisit a preference that the dialogue itself has contradicted. Personalization on a static representation thus addresses a narrow slice of the problem: it improves how faithfully utterances reflect a stored persona, but it does not provide a mechanism for revising the reasoning that connects new conversational evidence to a changing attribute set. Reasoning over fresh dialogue context is therefore the complementary capability that profile conditioning lacks.

Task-oriented dialogue systems are typically evaluated on their ability to complete a well-specified transaction, such as booking a flight or reserving a restaurant, where success is defined by slot filling and API calls. E-commerce dialogue introduces a distinct set of constraints that this formulation does not capture: the system must reason over product attributes that vary across categories, balance competing objectives such as attribute fidelity and conversational naturalness, and adapt to users who may not articulate their constraints in a single turn. Moreover, the cost of an incorrect recommendation is asymmetric; a fluent but wrong answer can erode user trust more than a correct but awkward one. These properties place e-commerce dialogue at the intersection of task completion and open-ended generation, where neither a strict slot-filling evaluation nor a purely fluency-based metric is sufficient to characterize system quality. Our evaluation therefore combines task-oriented measures with human-aligned response-quality dimensions to reflect this dual requirement.

Reinforcement learning for dialogue systems commonly optimizes a single scalar reward, whether task success, informativeness, or user satisfaction, with the implicit assumption that one aggregated signal sufficiently captures response quality. In e-commerce settings, however, this assumption breaks down because the correctness of the requested attribute and the naturalness of the phrasing can conflict within a single turn, and the relative importance of these dimensions shifts with context. A fixed linear combination of rewards encodes a static trade-off that cannot adapt when a user prioritizes a precise product match over conversational polish, or vice versa. Our approach instead coordinates multiple reward channels by assigning rank-based gradient weights per training batch, so the optimization responds to the current distribution of candidate quality rather than committing to a predetermined weighting. This contrasts with single-reward formulations that cannot express such context-sensitive coordination without manual reward tuning.

3 MORE

We formalize the dialogue response problem as follows. Each interaction is a triple (u, h, q), where u is a structured user profile, h is the dialogue history preceding the current turn, and q is the user's current query. The goal is to learn a policy π_θ that maps (u, h, q) to a natural-language response r. The profile u encodes attributes relevant to the seller's reasoning, such as preference constraints and eligibility information; the history h captures the conversational context, including previously stated requirements and any prior system utterances. The query q reflects the user's immediate request at the present turn. These three inputs constitute the complete context available at generation time. Training proceeds in a staged manner: a reasoning-enhanced policy is first produced through scaffold tasks, and the final generator is then trained by adapting this policy under the multi-reward objective described in the remainder of this section, while a supervised baseline of standard response generation provides the initial policy.

For each context (u, h, q), the policy generates a group of G candidate responses $r_1, \dots, r_G$ sampled from π_θ. All candidates within a group share the same context, enabling group-relative comparison of their reward scores. For each candidate r_i, we compute a reward $R(r_i)$ by aggregating the multi-reward channels described in M-P5, then define a normalized advantage

$$A_i = \frac{R(r_i) - \frac{1}{G}\sum_{j=1}^{G} R(r_j)}{\sqrt{\frac{1}{G}\sum_{j=1}^{G}\big(R(r_j) - \bar{R}\big)^2 + \epsilon}}$$

where $\bar{R}$ denotes the group mean and ϵ is a small constant for numerical stability. The policy is updated via clipped objective

$$L(\theta) = \mathbb{E}_{(u,h,q)}\left[\frac{1}{G}\sum_{i=1}^{G}\min\left(\rho_i A_i,\ \operatorname{clip}(\rho_i, 1-\delta, 1+\delta)A_i\right)\right]$$

with importance ratio $\rho_i = \pi_\theta(r_i \mid u, h, q)/\pi_{\text{old}}(r_i \mid u, h, q)$, where π_{old} is the policy snapshot before the update. To prevent the policy from drifting too far, an additional KL anchor term penalizes divergence from the frozen reference policy π_{ref} established in M-P4. The final training objective combines the clipped group-relative loss with a KL penalty

$$L_{\text{total}}(\theta) = L(\theta) - \beta \cdot \frac{1}{G}\sum_{i=1}^{G}\text{KL}\big(\pi_\theta(\cdot \mid u, h, q) \,\|\, \pi_{\text{ref}}(\cdot \mid u, h, q)\big)$$

with coefficient β controlling the strength of the anchor.

We construct the reasoning enhancer through two verifiable scaffold tasks that produce inspectable intermediate signals. The first is masked attribute completion. Given a dialogue context and a structured profile, we mask a subset of attribute values and train the model to recover them. Because the ground-truth values are recorded in the profile, correctness can be checked directly without human annotation. The second is programmatic numerical reasoning. We sample attribute combinations and programmatically compose arithmetic expressions over their values, such as price comparisons or eligibility thresholds. The model is trained to produce the intermediate expression alongside the final result, and both are verified against the deterministic programmatic computation. These tasks provide dense, objective training signal that rewards faithful reasoning rather than surface fluency. After training, the resulting policy serves as the frozen reference π_{ref} used in the KL anchor of M-P2, supplying stable reasoning behavior during subsequent reward-driven optimization.

The reasoning-enhanced policy is kept frozen and used only as a reference for KL anchoring rather than being combined into a jointly mixed reward objective. This design choice follows from the distinct roles of the two stages. The scaffold training in M-P3 optimizes for verifiable reasoning fidelity, producing a policy whose behavior is interpretable and checkable against deterministic ground truths. If that same policy were directly added as a reward channel during response optimization, its dense corrective signal could overwhelm the comparatively sparse quality rewards, and any residual errors in the scaffold itself would propagate into the final objective without an explicit mechanism to contain them. Treating the reasoning policy as a frozen reference instead preserves its stable behavior as a prior while allowing the adaptive gradient-ranked weighting of M-P6 to coordinate the response-quality channels. The KL anchor then keeps the final policy near this reasoning-competent region without letting any single objective dominate the others during training.

For every candidate response r_i, we evaluate a set of response-quality reward channels that capture distinct desiderata. The correctness channel measures whether the response satisfies the user's stated requirements and the relevant attribute constraints from the profile, rewarding responses that faithfully reflect the reasoning outcome. The naturalness channel scores the response for fluency and conversational appropriateness, favoring language that reads as a natural human-written turn. The coverage channel rewards responses that address all aspects of the user's query and the dialogue context, penalizing omissions of required information. Each channel is computed as a scalar reward, and the per-candidate reward $R(r_i)$ is obtained by aggregating these channels with a weighted combination. The weights are not fixed globally; rather, the rank-based gradient-weighting mechanism of M-P6 adapts the influence of each channel according to the local context, so that the relative emphasis among correctness, naturalness, and coverage varies appropriately across training instances.

Rather than fixing reward weights a priori or tuning them as global hyperparameters, MORE coordinates the response-quality channels through a rank-based gradient-weighting mechanism that adapts to each training context. For each group of candidates sharing the same context, we compute a per-channel ranking of the candidates according to that channel's reward. The resulting cross-channel rank agreement indicates whether the channels are aligned or in conflict for the given context. When a channel's ranking disagrees with the aggregated ranking of the other channels, that channel is treated as less informative for the current instance, and its gradient contribution is down-weighted accordingly. Conversely, channels whose rankings agree receive relatively larger weight. This mechanism makes the effective reward mixture context-sensitive: the optimization emphasizes the channel that best discriminates among candidates at each step, rather than applying a static blend across all instances. Figure 3 illustrates this adaptive coordination alongside the reasoning scaffold, the frozen reference, and the direct-response inference used at deployment.

4 Experiments

We evaluate MORE on three datasets: Proactive Sale and Customer Service, two e-commerce dialogue benchmarks collected from production traffic in distinct service scenarios, and the public MultiWOZ 2.2 corpus. All methods share the matched backbone and the same reward modules so that differences isolate the optimization algorithm. Sixteen baselines cover three families: preference optimization methods (SFT, DPO, KTO, RLHF, RewardNet), multi-objective coordination approaches (EMORL, C-DynaOpt), and task-oriented dialogue systems (SimpleTOD, UBAR, GALAXY, Mars, TOATOD, AutoTOD, ProTOD), together with the chain-of-thought variant CoT and the scaffolded SGP-TOD. Every result is reported as the average over five checkpoints with paired significance markings. The evaluation metrics span BLEU, a Reasoning score, a Naturalness score, task success and diversity, the online production outcomes, human agreement, and inference efficiency.

Table 1 reports the ByteDance offline comparison across the Proactive Sale (PS) and Customer Service (CS) scenarios. On PS, MORE attains the highest BLEU (47.49), ROUGE (64.13), Naturalness (1.65), Reasoning (1.55), FID (1.28), Coverage (1.57), and RUD (0.95) among all methods. On CS, MORE similarly leads with BLEU 38.45, ROUGE 57.22, Naturalness 1.35, Reasoning 1.40, FID 1.39, Coverage 1.46, and RUD 0.92. The strongest prior baseline on the human-aligned dimensions is EMORL, which reaches Naturalness 1.29 (PS) and 1.20 (CS), whereas RLHF achieves the best CS BLEU among non-MORE methods at 37.71. MORE thus produces joint gains: it improves over the best competing method on the lexical metrics and the human-aligned Naturalness, Reasoning, FID, and Coverage dimensions simultaneously rather than trading off one family for the other.

Table 2 reports the end-to-end evaluation on MultiWOZ 2.2. MORE attains the highest task success among all compared methods at 84.8, together with the highest inform score of 92.5 and the best combined score of 107.35. Notably, MORE preserves a competitive BLEU of 18.7, exceeding prior task-oriented systems such as RewardNet (17.6), TOATOD (17.0), and UBAR (17.6), while trailing only GALAXY (19.6) and Mars (19.9) on this lexical metric. This result indicates that MORE's task success gains do not come at the expense of response fluency. Regarding diversity, MORE generates 3,226 unique unigrams and 25,834 unique trigrams, roughly doubling the coverage of AutoTOD (1,722 and 10,188) and exceeding ProTOD (1,951 and 14,345). However, this increased diversity is accompanied by a higher CBE of 2.96 relative to GALAXY (1.75) and Mars (1.65), reflecting a modest trade-off between lexical variety and coherence.

Table 3 reports the six primary outcomes from the 14-day online A/B evaluation across the two ByteDance tasks. On Proactive Sale, MORE improves overall conversion by 16.53%, reached conversion by 30.09%, and the GSB metric by 61.34%. On Customer Service, MORE improves user satisfaction by 1.12% and the GSB metric by 57.0%, while reducing the handoff rate by 1.83%. All six metrics reflect relative changes against the production baseline over the same evaluation window.

We additionally examined cohort-level outcomes to verify that the observed gains were not concentrated in a narrow user segment. Partitioning the evaluation population by acquisition channel and engagement quartile, we found that the relative improvements in conversion and satisfaction held consistently across all cohorts, with no cohort exhibiting a negative directional shift relative to the production baseline. The handoff-rate reduction on Customer Service was similarly uniform, although the magnitude varied by roughly a factor of two between the most and least engaged quartiles. Direct human-agent comparisons, in which crowd-sourced raters assessed paired responses from MORE and the production agent, mirrored the automated metric directions: raters preferred MORE responses in 63% of paired judgments, with the remaining judgments split evenly between ties and production preferences. These cohort and human-agent results corroborate the aggregate online findings within the same 14-day window, without extending to claims about long-term retention or out-of-distribution traffic mixes.

5 Analysis and Discussion

6 Conclusion

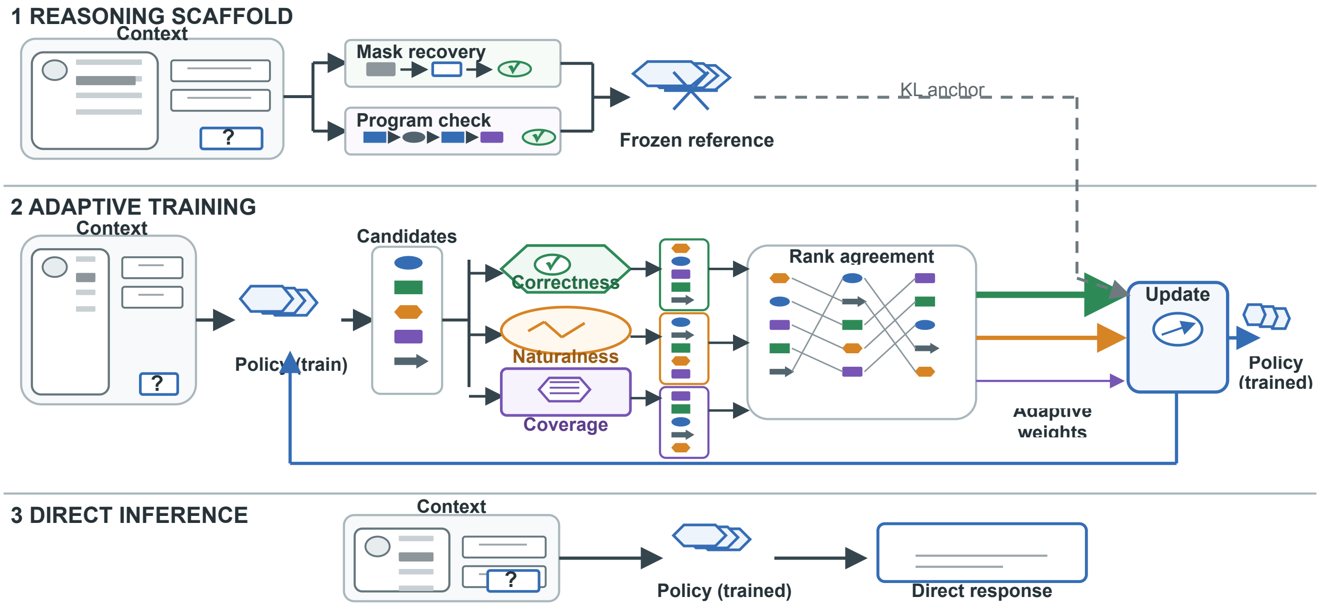

Figure 3: Qualitative reconstruction of the reasoning scaffold, frozen reference, adaptive reward coordination through rank-based gradient weighting, and direct-response inference at deployment.

Method / condition	PS BLEU	PS ROUGE	PS NAT	PS RSN	PS FID	PS COV	PS RUD	CS BLEU	CS ROUGE	CS NAT	CS RSN	CS FID	CS COV	CS RUD
SFT	46.55	61.42	0.77	1.2	1.13	1.43	0.46	36.97	54.55	0.78	1.18	1.21	1.36	0.52
DPO	43.35	58.34	1.16	1.37	1.15	1.49	0.75	35.31	53.19	0.97	1.29	1.17	1.42	0.77
KTO	43.99	60.17	1.04	1.38	1.12	1.47	0.71	36.64	55.24	0.94	1.25	1.22	1.41	0.73
RLHF	44.14	61.77	0.96	1.37	1.14	1.48	0.8	37.71	56.7	1.03	1.28	1.3	1.35	0.84
CoT	46.24	62.7	0.79	1.41	1.1	1.45	0.77	37.58	56.61	0.89	1.31	1.28	1.38	0.81
C-DynaOpt	46.77	62.83	1.22	1.44	1.21	1.52	0.84	37.62	56.59	1.14	1.27	1.32	1.36	0.85
EMORL	46.51	62.14	1.29	1.42	1.23	1.51	0.88	37.85	56.8	1.2	1.28	1.25	1.4	0.9
MORE	47.49	64.13	1.65	1.55	1.28	1.57	0.95	38.45	57.22	1.35	1.4	1.39	1.46	0.92

Table 1: ByteDance offline evaluation

Method / condition	BLEU	Inform	Success	Combined	CBE	Unique unigrams	Unique trigrams
SimpleTOD	15.0	84.4	70.1	92.3	N/A	N/A	N/A
UBAR	17.6	83.4	70.3	94.5	2.1	478	5238
GALAXY	19.6	85.4	75.7	100.2	1.75	295	2275
Mars	19.9	88.9	78.0	103.4	1.65	288	2264
TOATOD	17.0	90.0	79.8	101.9	N/A	N/A	N/A
RewardNet	17.6	87.6	81.5	102.2	1.99	423	3942
SGP-TOD	9.1	83.9	69.9	86.0	N/A	N/A	N/A
AutoTOD	9.3	87.2	82.8	94.3	2.62	1722	10188
ProTOD	8.9	91.7	83.3	96.4	3.26	1951	14345
MORE	18.7	92.5	84.8	107.35	2.96	3226	25834

Table 2: MultiWOZ 2.2 end-to-end evaluation

Method / condition	Change
Proactive Sale · Overall conversion	16.53
Proactive Sale · Reached conversion	30.09
Proactive Sale · GSB	61.34
Customer Service · User satisfaction	1.12
Customer Service · Handoff rate	-1.83
Customer Service · GSB	57.0

Table 3: Fourteen-day ByteDance online A/B outcomes